# Integration by Substitution

## Section 1: Basic u-Substitution

**1** Evaluate $\int 2x(x^2+5)^4\,dx$.

**a.** $\frac{1}{5}(x^2+5)^5+C$

**b.** $\frac{1}{4}(x^2+5)^4+C$

**c.** $(x^2+5)^5+C$

**d.** $\frac{2}{5}(x^2+5)^5+C$

**2** Evaluate $\int x^2\sqrt{x^3+1}\,dx$.

**a.** $\frac{2}{9}(x^3+1)^{3/2}+C$

**b.** $\frac{2}{3}(x^3+1)^{3/2}+C$

**c.** $\frac{1}{3}(x^3+1)^{3/2}+C$

**d.** $\frac{3}{2}(x^3+1)^{1/2}+C$

**3** Evaluate $\int \frac{x}{x^2+9}\,dx$.

**a.** $\frac{1}{2}\ln(x^2+9)+C$

**b.** $\ln(x^2+9)+C$

**c.** $\frac{1}{3}\arctan(x/3)+C$

**d.** $\frac{1}{2(x^2+9)^2}+C$

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## Section 2: Trigonometric and Exponential Substitutions

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**4** Evaluate $\int \sin^3(x)\cos(x)\,dx$.

**a.** $\frac{1}{4}\sin^4(x) + C$

**b.** $\frac{1}{3}\sin^3(x) + C$

**c.** $-\frac{1}{4}\cos^4(x) + C$

**d.** $\sin^4(x) + C$

**5** Evaluate $\int \tan(x)\,dx$.

**a.** $\ln|\sec x| + C$

**b.** $-\ln|\sec x| + C$

**c.** $\sec^2(x) + C$

**d.** $\ln|\sin x| + C$

**6** Evaluate $\int e^{3x-1}\,dx$.

**a.** $\frac{1}{3}e^{3x-1} + C$

**b.** $3e^{3x-1} + C$

**c.** $e^{3x-1} + C$

**d.** $\frac{1}{3}e^{3x} + C$

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## Section 3: Definite Integrals with Limits Change

**7** Evaluate $\int_0^1 x(x^2+1)^3\,dx$.

**a.** 15/8

**b.** $16/8 = 2$

**c.** 7/8

**d.** 15/4

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**8** Evaluate $\int_0^{\sqrt{\pi}} x\sin(x^2)\,dx$.

**a.** 1

**b.** 0

**c.** 2

**d.** 1/2

**9** Evaluate $\int_1^e \frac{\ln x}{x}\,dx$.

**a.** 1/2

**b.** 1

**c.** 2

**d.** $1/e$

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## Section 4: Advanced Linear Substitutions

**10** Evaluate $\int x\sqrt{x+2}\,dx$.

**a.** $\frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C$

**b.** $\frac{2}{3}(x+2)^{3/2} + C$

**c.** $\frac{1}{5}(x+2)^{5/2} - \frac{2}{3}(x+2)^{3/2} + C$

**d.** $\frac{2}{5}x^{5/2} + C$

**11** Evaluate $\int \frac{x}{\sqrt{x+1}}\,dx$.

**a.** $\frac{2}{3}(x+1)^{3/2} - 2\sqrt{x+1} + C$

**b.** $\frac{1}{3}(x+1)^{3/2} - \sqrt{x+1} + C$

**c.** $2\sqrt{x+1} + C$

**d.** $\frac{2}{3}(x+1)^{3/2} + C$

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**12** Evaluate $\int \frac{e^{2x}}{e^x + 1}\, dx$.

**a.** $e^x - \ln(e^x + 1) + C$

**b.** $e^x + \ln(e^x + 1) + C$

**c.** $\ln(e^x + 1) + C$

**d.** $\frac{1}{2}e^{2x} + C$

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